Bye-Bye Big Kat

Jerry Pournelle

Big Kat goes off to college, and the flaky Zelda replaces him

I started this on the BART train under the San Francisco Bay. I was trying to write on Zebediah, my Zenith Z-181 portable, but I didn't get very much done.

It wasn't Zeb's fault. He is a bit heavy, and the case is slick enough that it will slip off your lap if you're not careful, but it is possible to hold the machine there; and that bright blue backlit screen is readable in any light conditions I've yet found. In my experience, two full-feature PC-DOS portables are worth considering: the Z-181 and the Toshiba T3100 with its hard disk. The Toshiba isn't a laptop, but the Z-181 is supposed to be. It runs for hours on batteries. The screen is large enough. But I simply couldn't write on that BART train.

The problem wasn't the computer, but the American Tourister case. I found it impossible to leave the machine in the case, and nearly as difficult to take the machine out and close it up. There's almost no way to keep disks in that case once it's been opened: the thing is optimally designed to drip its contents onto the floor. Since the Z-181 runs only a few hours—three to five, depending on how much disk access you do—between charges, I generally want the somewhat heavy and badly designed power supply in the case with the Z-181, and there's no separate compartment for it. The cords very much get in the way.

In fact, there aren't any truly separate compartments for *anything* in that case. There's a flap, but it's made of thin cloth and holds nothing. The result is that the only stiffening is provided by the computer itself, and once you remove that, all the disks, documents, templates for WordPerfect, small tools, cables, and other stuff that you'd like to keep with a full-featured portable are mixed together into a random stew—or else the stuff falls out onto the floor, as it did with me. I actually missed my BART stop because I was trying to retrieve disks that had dripped from the Z-181's case.

Apparently Zenith put all its design skill into the Z-181 itself and ordered the case and power supply as afterthoughts. If the Z-181 sells as well as it ought to, someone will make a fortune selling well-designed cases to carry it in. While they're at it, they might design a lightweight handle to attach to the computer itself.

Communication by Alligator

It's a tendency in modern hotels. The phone system is set up so you don't have to talk to anyone to dial out; just dial 8, or 9, and then the number. The phones are almost all Touch-Tone, too, so it ought to be easy to connect them to modems. It seldom is easy, though, because the telephone instrument is hard-wired into the system. Since nearly all modems now expect you to use a modular plug, this presents a problem if it's midnight in the San Francisco Hilton and you desperately need to log on to BIX.

I was in that situation recently. I thought of two possible solutions. First, I carry a spare telephone cord. The *instrument* was hard-wired, but was the other end at the wall? Alas, that was well behind the bed, and short of moving a massive piece of furniture, I wasn't going to find out. (Incidentally, why are phones always at the bed and never at the worktable, even in hotels that say they cater to business travelers?)

The second solution was given to me by Dr. Robert Bussard, who sent me a modular plug that ends in four color-coded alligator clips. I'd carried it to many places, but I'd never been desperate enough to use it. This time I was.

With a certain amount of trepidation, I took the telephone apart with my Swiss army knife. The cover came off easily enough, and once inside the job was simple, barring the fact that the phone was now permanently off the hook. The wires from the wall plug terminated just inside the instrument and had the four colors all phone lines seem to have now. Two of them are still called "tip" and "ring," presumably carryover names from the days when phone plugs and jacks were larger. I don't know which is which, or indeed which two of the four lines are significant, so I used the alligator clips to connect similar colors of all four lines from my bit of phone cable and plugged the other end into the line port on the Z-181's modem.

Voilà! Everything worked fine. There was a bit of line noise, and after I was connected to BIX I managed to hang up the phone instrument. I was still connected, and most of the noise went away. As I'd suspected, the telephone instrument was now totally irrelevant.

I don't suppose the Hilton would have approved, but in fact I had no problems, either with BIX or in getting the phone put back together. Let me hasten to add that I don't recommend that you try this.

Communications, Continued

Last month I mentioned the difficulties I had in getting the Z-181 to communicate files to Big Kat, the Kaypro 286i PC AT clone. The Zenith PCXFER program says "use a null modem," but in fact that's an insufficient instruction. What they mean by "null modem" is a cable of considerable complexity. I'll repeat the formula for those who missed last month's issue.

Take a standard 25-pin RS-232C cable and alter. Pin 1 goes straight through. Cross-connect (i.e., swap over) pins 2 and 3. Swap 4 and 5. Cross-connect 6 and

continued

Jerry Pournelle holds a doctorate in psychology and is a science fiction writer who also earns a comfortable living writing about computers present and future.

20. Pins 7 and 8 go straight through. Given all that, PCXFER worked fine to transfer files to and from Lucy Van Pelt, our genuine IBM PC; and indeed, it works with any normal PC or XT. The program is a little slow, but it's automatic and understands wild-card instructions.

I couldn't get PCXFER and that cable to work with an AT. It turns out there was a very good reason: 25-pin connectors on AT-style machines, including the Kaypro, are parallel ports. Serial ports on an AT have only 9 pins.

Okay, I should have known that. It's even stated in Kaypro's documents, and now that I'm sensitized I've seen it elsewhere. The fact is, I didn't know.

The IBM PC has two 25-pin ports, one serial, one parallel. The Z-181 has two 25-pin ports (actually three), and one is the serial port. The Kaypro 286i AT has two 25-pin connectors on the back. One is on the Hercules Color Card video board and is clearly labeled parallel. It ran the printer fine. The other is, of course, on the I/O card that also contains a 9-pin port. For not very intelligent reasons, I assumed those were two different serial ports, or possibly two different configurations of the same serial port. The 9-pin port was used to connect the Logitech mouse, and it worked fine. The 25-pin port wouldn't work with PCXFER, even with the snazzy "null-modem" cable I'd made to Zenith's specs.

It never would work, of course, because the 25-pin port is an extra parallel port and can even be connected to another parallel device, provided only that both devices aren't turned on at the same time.

I found this out from BIX correspondents shortly after filing last month's column. It says a lot for the wretched state of "standards" in this industry that so few were surprised by my problems that it took a week before someone noticed I was trying to do the impossible. People *expect* serial communications problems.

Alas, the Zenith PCXFER documents don't give any clues about cabling to a 9-pin connector. I have a couple of cables that do have a 25-pin port on one end and a 9-pin port on the other, and using those and WireTap (an inexpensive breakout box; see last month's column) I experimented for an hour, but I never did get the Z-181 to talk to the Kaypro or to a Zenith Z-248 AT clone for that matter.

Eventually I ran out of time. I used PCXFER to send my files from the Z-181 into Lucy Van Pelt, then used the CompuPro network to transfer those files to the big CompuPro, and from that to the Kaypro. It meant one more copy command, but nothing else special, so except for

continued

stringing the network wires it wasn't a lot harder than doing it direct.

I've found another use for the CompuPro network. All my AT machines have 96-tracks-per-inch "quad-density" disk drives. Those drives *read* normal PC and XT disks fine; but when they write files, the probability that a 48-tpi "normal" PC drive will be able to read them is very low. The network solved the problem nicely: we use it to transfer the file to a PC with 48-tpi drives and let it write the file. Now anything can read it.

Enter Zelda

For more than a year, the main PC-DOS machine at Chaos Manor has been Big Kat. He's been so good, in fact, that I've turned down numerous offers of competing AT machines.

He's still working fine; but eventually it was time for a change, if only because I can't go on writing about the same machine forever. When Zenith asked me to look at the Z-248 AT clone, it wasn't a difficult decision. It's the standard machine used at the service academies, and I'm rather fond of Zenith anyway. Their stuff is reliable, and I like their ROM monitor. It took Zenith a while to get one of the machines to me, and until they did I kept Big Kat on-line; but eventually the Z-248 arrived.

First things first. Get the new machine running. That was no problem. When I turned it on, it said "not a bootable partition." When in doubt, read the instructions, and five minutes later I knew how to boot the machine with its floppies and partition the 30-megabyte hard disk. Another five minutes and the hard disk was formatted and set up with command files to be the boot disk.

There was one potential problem. The Z-248 came with an enhanced graphics adapter color board and Zenith's ZVM-1380 13-inch EGA color monitor. I was a little worried that text on the screen might not be big enough for me to be comfortable with. Also, about once every dozen times when I booted the system, the screen images were broken and fuzzy. The condition corrected itself on rebooting, but it was annoying.

It didn't seem fatal. I put it down to something being jarred in shipping and figured that at worst Zenith would send me a new video board.

Now I was left with the question of what to do with Big Kat, a thoroughly debugged and operational AT clone who has served me well for more than a year. In fact, the decision was simple enough. Frank, my #2 son, is studying business and finance at USC and needs a computer. PCs are fine, but ATs are better; with permission from Kaypro, Big Kat would become Frank's machine.

Since Big Kat had been the main PC-DOS machine here, he had a lot of files I'd need on the Zenith. On the other hand, I have a lot of programs I wasn't about to send down to USC. In some cases, I legally own only one copy. Most publishers probably wouldn't mind if I lend my son a copy, but I won't do that without permission. Others are prerelease. And, of course, some are games: college students get enough distractions without my furnishing more. I wanted to keep those files for me, but they'd sure have to be removed from Big Kat.

The simple solution would have been to use Fastback: back up all of Big Kat's files, then restore them on the Z-248. I could then go through Big Kat's files and delete as needed. I didn't think of that. I suppose it's partly because Big Kat had a lot of files that ought to have been deleted long ago, and this was a good opportunity to clean things up. Anyway, I used the network.

Networking on My Own

I had an extra CompuPro ARCNET PC board. Since the Zenith was the Z-248, it seemed reasonable to address it as node 48 (Big Kat had already been designated as 28). Of course, that would be *hexadecimal* 48, since that's how you specify node numbers on the CompuPro network; but I have SideKick, which has a desk accessory calculator that translates hexadecimal to decimal to binary, so surely it should be no problem. I opened the CompuPro ARCNET PC documents with considerable hope.

Hah. Five minutes later I was hopelessly confused. Eventually I called CompuPro and asked for technical support. I got a young chap who seemed to know what he was doing. He chattered gaily about least significant bits and the like and told me how to set the paddles—the individual switches—on SW-2, the eight-paddle DIP switch that tells the CompuPro ARCNET PC board what node it is. I set them and hit Ctrl-Alt-Del.

Nothing happened, of course. You first have to turn off the machine before the ARCNET PC card can read its own switch settings. Try again.

Unfortunately, that didn't work either. Back to phone CompuPro. This time I got Len Ott, CompuPro's very practical software guru. Len's solution was simple: use DEBUG to see what number the ARCNET PC board thought it was reading. A minute later we knew it wasn't reading 48 hexadecimal. Back to the DIP switch and exactly reverse each paddle. Turn off the machine. Fire it up again. Voilà! Everything worked fine.

continued

As with most such problems, what was needed here was *examples*. In this case, it needed some drawings of switch settings and their corresponding numbers, plus a couple of tables. Fortunately, CompuPro has hired an editorial consultant to revise the documents. I have some new data for her attention.

After that the network worked like a charm. Of course, DOS is too stupid to let me use the network in a single operation; that is, you can't, at the Zenith, issue a command that will bring you a file from the Kaypro, nor can you, from the Kaypro, send a file to the Zenith. Instead, you have to use the Kaypro to send a file to, say, the Golem, my big CompuPro 286/Z80 machine, then go to the Zenith, set up the directory structure you like, and call in the files from the Golem.

To do this, I told both the Kaypro and the Z-248 that they had a local drive called M, which was actually the M-Drive/H (RAM disk) on the Golem. (The network lets you name any drive on any machine any arbitrary thing you like, as long as it isn't something you already have. For example, I could not call a networked drive C if I already had a hard-wired drive named C attached to that terminal.) This made things go faster.

Then I noticed that my newly transferred files didn't have date and time stamps. It took a second before I realized why. The Golem's M drive is formatted as Concurrent CP/M media, and while DOS files—including command files—are stored intact, and work fine, the date and time stamps were lost on the way through. That problem was cured by using the C logical drive partition on the Golem; C is configured as MS-DOS media.

After that we had no problems. I experimented with PC Sweep, a couple of other shareware file manager programs, and various other stuff, with no surprises. PC Sweep running on the Z-248 is perfectly willing to believe that the M drive is nothing special and will tag files for transfer or deletion; while the Golem is smart enough to accept commands through the network from the Zenith. In a couple of hours, I had reorganized and transferred all my files.

Norton's Wipedisk

Once I'd done all that, there was still a problem. I'd erased files from Big Kat, but it would still be possible for someone down at USC to recover them; and while I trust my son, I didn't want him to have to put unusually severe security restrictions on access to the machine.

One solution might be to use Fastback to record all the data, then format the hard disk. Alas, I have a couple of utility programs that swear they can recover data even from a disk reformatted under DOS, and while I wasn't sending those programs with Frank, I make no doubt there are plenty of USC students who either have them or could write their own. Surely there's a way to permanently erase files.

There are lots of PC-DOS utility programs, some in the public domain, some shareware; so many that it's impossible to keep track of them all. I've found by and large, though, that the latest edition of The Norton Utilities will solve at least 90 percent of the DOS problems I run into. Norton's programs are well documented, both on how to use them and what they do. It seemed reasonable to look into the Norton manual to see if there was help for the current situation.

There was. Norton's Wipedisk doesn't just erase files, it overwrites them. I'd never used it before, so I ran Fastback to get backup copies of all the files we'd just transferred, then invoked Norton's Wipedisk with the /E option, which, according to the manual, "specifies that only the

continued

erased or unused data space will be wiped; this saves current data, but wipes erased data."

I am pleased to report that it does just what it says. It's not particularly fast: on a 20-megabyte hard disk about half full, it took most of dinnertime. On the other hand, under the /E option it runs unattended, and when it was finished, I couldn't find a program that would recover any of the wiped data.

I've said this before: if you do much work with DOS, you should have a copy of The Norton Utilities. There may be a better public domain version of every program in the Norton set, but you'll sure spend a lot of time and effort collecting them all, and even then you'll have to sweat some pretty wretched documents. The Norton Utilities are well worth the price.

Cheetah

The next thing was to install the Cheetah Combo, a no-wait-state memory-expansion board for the PC AT. The Cheetah Combo is faster than the built-in memory on most IBM PC AT machines, which means that with a lot of ATs it's a good idea to remove memory the machine comes with and let the Cheetah memory replace it. That's easy to do: Cheetah's documents are clear, and they give you a program that shows you just how to set the switches.

The Zenith, however, is faster than most AT clones. It came with 500K bytes of no-wait-state memory. I set the Cheetah Combo switches to tell it that, and instantly I had a 640K-byte system with an unknown amount of expanded memory. Next thing was to tell the Z-248 about its new memory. That turns out to be easy to do, since unlike the Kaypro, the Zenith tells *you* how much conventional and expanded memory it has. You then have to go into the ROM monitor and change the setup. (I don't know why the machine doesn't just tell itself about the changes.)

The only use that I've found for expanded—as opposed to extended—memory is as RAM disk. Installing that is simple enough. Just use the VDISK.SYS that comes with PC-DOS. In my case, on advice from Cheetah's Gene Sumrall, I put `DEVICE = VDISK.SYS 4096 256 256/E` into the CONFIG.SYS file. That tells the RAM disk to grab everything it can (up to 4096K bytes), format it into 256K-byte sectors, and leave room for 256 entries in the directory. If I were short of memory, I might decide to leave room for only 128 entries.

Once that's done you have to reboot. The first time I tried it didn't work: I had VDISK.SYS in a subdirectory, and the PC looks for CONFIG.SYS in the root directory before it reads the PATH instructions that can tell it to look at subdirectories. Moving VDISK.SYS to the root directory fixed things, and at the next reboot I had 640K bytes of main memory and a 2.48-megabyte RAM disk labeled drive D.

The Cheetah Combo comes with some interesting software. One program will force your programs to use high memory before low to make efficient use of the superfast Cheetah no-wait memory. Because the Z-248 is already plenty speedy, I didn't bother with that except to test it out. It works, but I saw no real speed improvement. With another machine I probably would have.

Above Disc

RAM disk is useful, but my problem is that I like memory-resident software. That eats memory, and neither expanded memory nor RAM disk is a bit of help. What I need is "extended" or "smart" memory, generally abbreviated as EMS. I called Cheetah to see if they had anything that would help and learned about

continued

Above Disc virtual memory emulator.

Above Disc is a memory-resident program that uses about 50K bytes of regular memory. It uses that to control a specified portion of your RAM disk and turn that into the equivalent of extended or smart memory. The program swaps in and out of regular memory according to the rules of the EMS standard. In other words, it creates what the big computer people call virtual memory.

Installing Above Disc is simple, provided that you *read the entire manual* before starting. It isn't that long a manual, so this is no real problem, however there are a number of caveats about Above Disc, and you'd better be aware of them.

Once Above Disc was in place, I rebooted to see what it would do for Ready!. The result was very nice: Ready! took up 3K bytes of conventional memory and 160K bytes of the expanded memory created by Above Disc. Ready! also worked perfectly.

Unfortunately, Ready! is the only memory-resident program I'm addicted to that can make use of expanded memory. Philippe Kahn tells me he'll have an EMS version of SideKick Real Soon Now. However, the Above Disc scheme works with DESQview, Lotus 1-2-3, Symphony, Reflex 1.1, and other programs engineered to take advantage of EMS. The instructions explicitly say it will *not* work with Javelin.

Above Disc doesn't have to use RAM disk to do its thing: you can designate part of your hard disk, or even a floppy disk, as "expanded memory," and Above Disc will swap stuff in and out of a reserved area on that disk. Of course it won't be very fast, and if you remove the floppy at some point during the proceedings, you'll really mess things up; but it will work.

The DESQview people, and some others, tell me that Above Disc isn't *really* EMS-standard expanded memory, and I'm sure they're right; but it does quite a lot, and we haven't had any problems with it. The Cheetah Combo and Above Disc make a good combination.

Zelda Goes Mad

I'd had the Z-248 all weekend, and it was giving me no problems, so I sent Big Kat off to USC with Frank. Given the rugged construction of the Kaypro, I wasn't worried about it, and I was right: despite having a bunch of business students banging away on it, Big Kat is working fine.

Alas, that wasn't true of the Z-248. The instant Big Kat left the house, the Z-248 did odd things. First, her screen images began to dissolve, until there were no more than 25 percent of the pixels on the screen. English written in the roman alphabet has a lot of redundancy, so I could just make out words and commands; but it sure wasn't easy.

Resetting the machine would generally cure the screen problem, but it often would introduce a different difficulty: the Above Disc disk-swapping system would load itself and report that all was well, but then Ready!, loading itself on start-up, couldn't find it. Instead, about half the time, Ready! would load itself entirely into conventional memory, as if the Above Disc pseudo-expanded memory weren't there at all.

Then I had real problems getting the machine to work with the OmniTel modem. I've had that modem in Big Kat for two years with no glitch whatsoever, and I refused to believe it had gone sour—and indeed, rebooting would often cure that problem, too. After that I had problems with the ARCNET PC and even with the PATH command.

None of this happened every time, which was annoying. Computers aren't supposed to do things *sometimes*. It was time for some serious diagnostics.

continued

First things first. I took out the Cheetah Combo, ARCNET PC, and the OmniTel modem, leaving nothing but the original Zenith boards. That did no good, but then I hadn't expected it to. The machine still did odd things *sometimes*.

When I mentioned these problems on BIX, Barbara Clifford pointed out that the machine clearly should be named Zelda, after F. Scott Fitzgerald's very fast, and somewhat flaky, wife. Phone consultation with Wayne Rash, who did the formal (and very favorable) review of the Z-248 in the December 1986 issue of BYTE, produced a more technical diagnosis: "It sounds like it's all screwed up."

Waiting for Igor

I called Zenith for help. As luck would have it, the relevant Zenith officials weren't in their offices in Chicago. They were out here; to be exact, in Sante Fe Springs, not more than 20 miles from here. As a result, it took a full day to reach Glen Nelson and explain the problem.

"I don't know if it's the CPU or the video board or both," I said.

"Sounds like the machine was damaged in transportation," Nelson told me. "That's the identical machine Wayne Rash reviewed, and we tested it before we shipped it to you. Let me send you a new one."

That sounded reasonable, but for a test I asked that they send new CPU and video boards. Incidentally, that's one of Zenith's better features: the CPU and system memory are on cards just like everything else, making both repairs and upgrades much easier.

Glen promised to get the boards sent Federal Express, and I went back to trying to understand an increasingly inarticulate Zelda. She was deteriorating rapidly. Sometimes only 10 percent of the pixels would be displayed on-screen. She'd often lose all her path information. Sometimes she'd be unable to find the modem.

My wife watched over my shoulder during one of Zelda's bad spells. "Give up," she said.

"Good advice, but right now this is the only machine I can use to BIX."

"Good heavens. What will you do?"

"It's not so bad," I said. "They're sending a new brain for transplant."

"And when does Igor arrive?" Roberta asked.

"Tomorrow," I said.

Of course the new brain didn't come the next day, or the day after, so there we were, like Dr. Frankenstein waiting for his crazed assistant to come back with a new brain.

Eventually the boards arrived, with mixed results: the video problems went away, but the Above Disc program didn't consistently work. At that point I went off to a conference.

Zelda Cured

When I got back there was a new Z-248 waiting.

First things first. Boot it up. Partition the hard disk. Format it. Transfer DOS to it. Boot again. Everything worked fine. Time to transfer files.

This time I took the simple way: I used the new Fastback with no copy protection. I copied the floppy disk to my hard disk, then tried to run the program. That didn't work. Even though there's no copy protection, you must use the Install program. Once you've done that you can put the floppy away.

Fastback is quite easy to use, and pretty soon I had all my programs stored on 12 quad-density disks. The next thing was to transfer all those over to the new Zenith Z-248. I had a moment of panic at my first attempt: the Fastback Restore

continued

The bottom line is that I've been using Zelda for three weeks and the cover hasn't come off again. She's fast and reliable and the nicest AT I've come across.

program has several options on what to do about old files, and I'd chosen the "erase them" route. Since there weren't any files to erase, Fastback wouldn't do anything. Eventually I figured that out and changed options, after which Fastback roared away. In less than half an hour I'd not only transferred all the files, but made a backup copy I could keep.

At that point I had a decision. Zelda II seemed to be working fine. Did I dare trust her? After all, Zenith does make good equipment. Wayne Rash liked the Z-248 best of all the AT machines he tested.

Why not, thought I, and proceeded to install the Cheetah Combo, OmniTel modem, and CompuPro ARCNET PC board. Then I not only put the cover on, I screwed it down before turning it on.

Despite such arrogant temptation of fate, the Zenith came up perfectly. Ready! went into the Above Disc equivalent of EMS. SuperKey and SideKick installed fine. The CompuPro ARCNET PC worked perfectly.

The bottom line is that I've been using Zelda for three weeks and the cover hasn't come off again. She's fast, reliable, and quite the nicest AT I've come across.

EGA

As I mentioned, Zelda came with an EGA board and a Zenith 13-inch EGA color monitor. I had some worries about that: I normally wear bifocal glasses, and I don't care to sit close to a screen and tilt my head, yet if I get far enough away to read through the top part of my glasses, the letters are often too small. I'd solved the problem with Big Kat by having his Hercules board feed into a 19-inch Zenith high-resolution color monitor (which also doubled as my VCR monitor when I felt like watching a movie); but you can't feed an EGA into a CGA monitor. Zelda's EGA board produced sharp and crisp letters, but would they be large enough?

The answer turns out to be "almost." That is: I've set up Zelda's screen at eye level and about 30 inches from my nose, and I can indeed read what goes up on it. Not as well as I can read Zeke's 15-inch monochrome screen, but in fact well enough that if I didn't have Zeke I'd be willing to use Zelda to write my books. I'm still looking for a good 15-inch EGA monitor, and I think I have a line on one; and for that matter, it can't be long before Zenith produces 15-inch and even larger EGA monitors. I'm now thoroughly convinced that EGA will be the new standard for business use.

Why Use Zeke?

Back in 1977 I looked at all the available computer systems that could be used for business and serious writing. In those days S-100 systems dominated the micro industry, and it wasn't hard to decide what to get. I bought Zeke II, a CompuPro Z80 system. My partner Larry Niven, who knows little about computers and wants to know less, had our consultant Tony Pietsch build him a copy (actually, *two copies*) of what I got.

Neither of us has ever regretted that decision. We have written millions of words, including three best-selling novels, on those computers. They still work, and work well.

Unfortunately, they're not immortal. Tony Pietsch is moving to Santa Barbara. Spare parts are hard to come by.

There are other intimations of mortality. Last week, as is my practice, I packed Zeke—the world's least portable micro—into the trunk of my convertible and headed for a motel, there to get as much done on a novel—*Prince of Mercenaries*, a Falkenberg/CoDominium novel, for those interested—as I could in five days without telephone and mail interruptions. When I got there and set Zeke up, blue smoke poured out of his 8-inch disk-drive enclosure.

It was clearly panic time, but in fact I managed to stay calm. Proceed slowly. Take things apart. Unplug cables. Work logically. After about 10 minutes it was clear: something had happened to the power supply in the disk-drive enclosure. The drives themselves were all right and so was the computer.

A quick call to Priority One established that they had, in stock, 8-inch disk-drive enclosures with power supply at a reasonable price. I drove out, bought one, brought it back to the motel, and remounted the drives. Within two hours Zeke was humming along. Incidentally, as far as I can tell, transporting Zeke had caused a nut to work loose from one of the bolts holding the fan, and that nut had shaken into a spot where it could short

continued

something. Anyway, I'd lost little but time, and not a lot of that; but it does give pause to think. Suppose I couldn't find a new enclosure?

A few days later, I was talking with Larry and Marilyn Niven. Marilyn, unlike Larry, does know something about computers. After all, she took a degree in mathematics under Marvin Minsky at MIT. We all agreed that one day soon we'd have to replace our ancient Z80 machines with something else.

Given the way the world is going, that something else will likely be some kind of PC AT compatible with an EGA card. We'll also need a new text editor, one that's as nearly like WRITE as we can find; and that will probably be WordPerfect.

None of this is certain, but we've got WordPerfect 4.2 working with the CompuPro ARCNET PC, something that I couldn't do with 4.1. Now if we can just get oversize EGA monitors.

Still, things change like dreams in this micro world. At CES I was taken upstairs and shown the new Commodore Amiga 2000; the one I saw combined in the same box both a 68020 Amiga and an 8086 PCompatible; and the PC side of the box could take accelerator boards like the Orchid Turbo EGA. Alas, what they later announced as the 2000 wasn't what I saw; but Commodore potentially has a machine to compete with the Macintosh II if they'll just develop it properly.

Meanwhile, Cheetah has a 386 add-on that will turn a normal AT into about nine virtual machines, neatly solving the problem of what to do about memory-resident programs.

By the time you read this, Apple will have their Mac II, the open Macintosh, and IBM will have a new 386. Microsoft will have Advanced DOS Real Soon Now.

With developments like that going on, why change until you have to?

Winding Down

As usual, I started this column full of good intentions. I spent hours selecting software and *covered* my desk with really good stuff; and now, before I've got to much of it, I'm out of space.

I do want to say something about Russ Walter's amazing series, *The Secret Guide to Computers*, which is now available in three volumes, $8 each, from the author. If you need to understand computers and haven't had much luck at it; if you have to teach other people about computers; if you just want to read some good books about computers; then send a check for $24 to Russ Walter, 22 Ashland St., Somerville, MA 02144. Interested parties are invited to write for a brochure before ordering. I can't think you won't like the books, and even if you don't, you'll find someone to give them to who will. Highly recommended.

Then there were the MacWorld Exposition and the Seybold Conference on Desktop Communications. Both were worth attending. On the other hand, the most exciting announcement at the Seybold Conference was Appleshare, a very impressive networking program that links Macintoshes; and there's not a lot of point in talking about the Macintosh until Apple releases the open Mac. The delay between my writing this and your seeing it is, due to heroic efforts by the BYTE publishing people, getting shorter, but it's not *that* short.

There are two books of the month: *Flim-Flam* by James Randi (Prometheus Books, 700 East Amherst St., Buffalo, NY 14215) and *On the Frontiers of Science: Strange Machines You Can Build* by G. Harry Stine (Atheneum). James ("the amazing") Randi is a lifelong skeptic who constantly carries a check for $10,000 that he will pay to anyone who can show him a "paranormal" result he can't explain or duplicate with prestidigitation; while Harry Stine is a rocket engineer who doesn't exactly believe in odd things like dowsing, but who claims to have used such devices with consistent results and will tell you how to make and operate dowsing rods. I've met Randi, and I find his extreme skepticism irritating, while Harry Stine has been a close friend for years. One day I'll build and test some of the machines he describes. Meanwhile, if I had to bet money, I'd bet with Randi, but my heart's with Harry Stine's view.

The game of the month is Strategic Conquest Plus for the Macintosh. It used up far more of my time than I'll admit to *anyone* and is quite simply the best strategic microcomputer war game I've ever encountered. I've managed to win at level 13, but just barely. Now I'm at 14, and in the fight of my life. Highly recommended so long as you understand that it's *really* addicting.

Last-minute report: Big Kat is doing fine at USC. It's a great machine for a student. Now, with any luck, I'll get to that mess of stuff on my desk. ■

Items Discussed

Above Disc $99.95
Tele-Ware West
909 Electric Ave., Suite 202
Seal Beach, CA 90740
(800) 654-5301

ARCNET PC $550
CompuPro
26538 Danti Court
Hayward, CA 94545
(415) 786-0909

Cheetah Combo $395
Cheetah International Inc.
107 Community Blvd., Suite 5
Longview, TX 75602
(800) 243-3824

Fastback $179
Fifth Generation Systems
909 Electric Ave., Suite 308
Seal Beach, CA 90740
(213) 493-4483

Kaypro 286i PC AT $2995
Kaypro Corporation
533 Stevens Ave.
Solana Beach, CA 92075
(619) 481-4300

Ready! $99.95
Living Videotext
2432 Charleston Rd.
Mountain View, CA 94043
(415) 964-6300

Strategic Conquest Plus $59.95
PBI Software Inc.
1163 Triton Dr.
Foster City, CA 94404
(415) 349-8765

The Norton Utilities 4.0 .... $100
Peter Norton Computing
2210 Wilshire Blvd., Suite 186
Santa Monica, CA 90403
(213) 453-2361

Z-248 .... $4999
Zenith Data Systems
1000 Milwaukee Ave.
Glenview, IL 60025
(312) 699-4848

Jerry Pournelle welcomes readers' comments and opinions. Send a self-addressed, stamped envelope to Jerry Pournelle, c/o BYTE, One Phoenix Mill Lane, Peterborough, NH 03458. Please put your address on the letter as well as on the envelope. Due to the high volume of letters, Jerry cannot guarantee a personal reply.